CHAPTER 7 TECHNIQUES OF INTEGRATION

7.1 Integration by Parts

- **Integration by Parts**

$$\int udv = uv - \int vdu$$

$$\int_a^b u(x)v'(x)dx = [u(x)v(x)]_a^b - \int_a^b v(x)u'(x)dx$$

Example 7.1. Find $\int x \cos x dx$.

let $u = x$ $\quad dv = \cos x\, dx$

$du = dx$ $\quad v = \sin x$

$$\int x \cos x\, dx = x \sin x - \int \sin x\, dx$$

$$= x \sin x + \cos x + C$$

Example 7.2. Find $\int_1^2 \ln x dx$.

let $u = \ln x$ $\quad dv = dx$

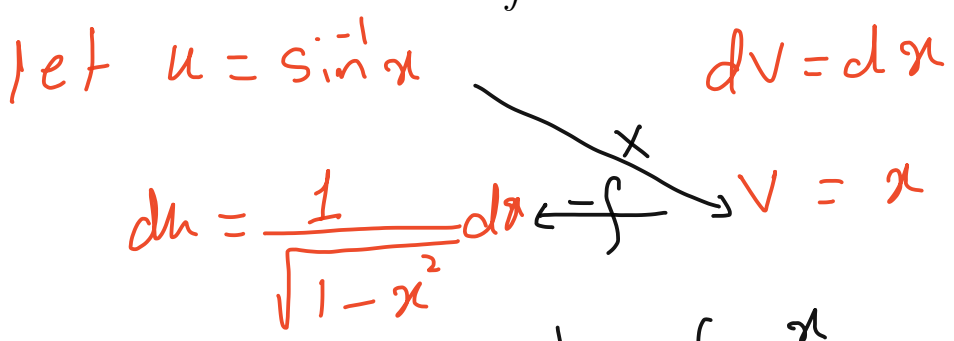
$du = \frac{1}{x} dx$ $\quad v = x$

$$\int_1^2 \ln x\, dx = \left[x \ln x\right]_1^2 - \int_1^2 x \left(\frac{1}{x}\right) dx = \left[x \ln x\right]_1^2 - \int_1^2 dx$$

$$= \left[x \ln x - x\right]_1^2 = \left[2 \ln 2 - 2\right] - \left[1 \ln 1 - 1\right]$$

$$= 2 \ln 2 - 2 + 1 = 2 \ln 2 - 1$$

Example 7.3. Find $\int \sin^{-1} x dx$.



$$\int \sin^{-1} x \, dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

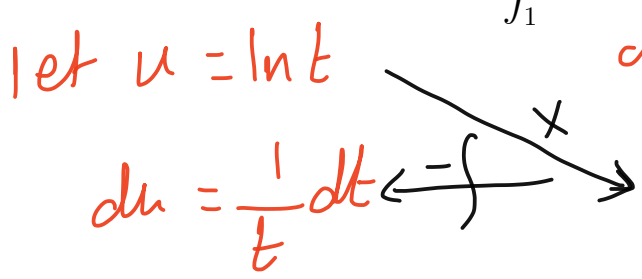
let $u = 1 - x^2$

$du = -2x \, dx \iff x \, dx = -\frac{1}{2} du$

$$\int \sin^{-1} x \, dx = x \sin^{-1} x - \int \frac{-\frac{1}{2}}{\sqrt{u}} du = x \sin^{-1} x + \frac{1}{2} \int u^{-\frac{1}{2}} du$$

$$= x \sin^{-1} x + \frac{1}{2} (2) u^{\frac{1}{2}} + C = x \sin^{-1} x + (1-x^2)^{\frac{1}{2}} + C$$

Example 7.4. Find $\int_1^2 t^6 \ln t dt$.



$$\int_1^2 t^6 \ln t \, dt = \left[\frac{1}{7} t^7 \ln t\right]_1^2 - \int_1^2 \frac{1}{7} t^6 dt$$

$$= \left[\frac{1}{7} t^7 \ln t - \frac{1}{49} t^7\right]_1^2$$

$$= \frac{128}{7} \ln 2 - \frac{127}{49}$$

Example 7.5. Find $\int x^2 \sin x dx$.

let $u = x^2$ $\quad dV = \sin x\, dx$

$du = 2x\, dx$ $\quad V = -\cos x$

$$\int x^2 \sin x\, dx = -x^2 \cos x + \int 2x \cos x\, dx$$

let $u = 2x$ $\quad dV = \cos x\, dx$

$du = 2\, dx$ $\quad V = \sin x$

$$\int x^2 \sin x\, dx = -x^2 \cos x + 2x \sin x - \int 2 \sin x\, dx$$
$$= -x^2 \cos x + 2x \sin x + 2\cos x + C$$

Example 7.6. Find $\int e^x \sin x dx$.

let $u = e^x$ $\quad dV = \sin x\, dx$ *

$du = e^x dx$ $\quad V = -\cos x$

$$\int e^x \sin x\, dx = -e^x \cos x + \int e^x \cos x\, dx$$

let $u = e^x$ $\quad dV = \cos x\, dx$

$du = e^x dx$ $\quad V = \sin x$

$$\int e^x \sin x\, dx = -e^x \cos x + e^x \sin x - \int e^x \sin x\, dx$$

$$2\int e^x \sin x\, dx = -e^x \cos x + e^x \sin x$$

$$\int e^x \sin x\, dx = -\frac{1}{2} e^x \cos x + \frac{1}{2} e^x \sin x + C = \frac{1}{2} e^x (\sin x - \cos x) + C$$

Example 7.7. Derive a reduction formula for $\int \sin^n xdx$. Hence, evaluate $\int_0^{\frac{\pi}{2}} \sin^8 xdx$.

Exercises Page 476 3, 5, 7, 9, 11, 13, 15, 17, 19, 23, 27, 29, 33, 35, 37, 41, 47, 49, 51, 53, 61, 63, 65.

7.2 Some Trigonometric Integrals

Example 7.8. Find $\int \sin^5 x dx$.

Example 7.9. Find $\int \cos^4 x dx$.

Example 7.10. Find $\int \frac{\sin^3 x}{\cos^4 x} dx$.

Example 7.11. Find $\int \sin^2 x \cos^4 x dx$.

- **Some Trigonometric Identities**

(a) $\sin mx \cos nx = \frac{1}{2}[\sin(m+n)x + \sin(m-n)x]$

(b) $\sin mx \sin nx = -\frac{1}{2}[\cos(m+n)x - \cos(m-n)x]$

(c) $\cos mx \cos nx = \frac{1}{2}[\cos(m+n)x + \cos(m-n)x]$

Example 7.12. Find $\int \sin 2x \cos 3x dx$.

Example 7.13. If m and n are positive integers, show that

$$\int_{-\pi}^{\pi} \sin mx \sin nx dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

Example 7.14. If m and n are positive integers, find

$$\int_{-L}^{L} \sin \frac{\pi mx}{L} \sin \frac{\pi nx}{L} dx$$

Example 7.15. Find $\int \tan^5 x dx$.

Example 7.16. Find $\int \cot^4 x dx$.

Example 7.17. Find $\int \sec x \tan^3 x dx$.

Example 7.18. Find $\int \tan^6 x \sec^4 x dx$.

Strategy for Evaluating $\int \sin^m x \cos^n x\, dx$

(a) If the power of cosine is odd ($n = 2k + 1$), save one cosine factor and use $\cos^2 x = 1 - \sin^2 x$ to express the remaining factors in terms of sine:

$$\int \sin^m x \cos^{2k+1} x\, dx = \int \sin^m x\, (\cos^2 x)^k \cos x\, dx$$
$$= \int \sin^m x\, (1 - \sin^2 x)^k \cos x\, dx$$

Then substitute $u = \sin x$.

(b) If the power of sine is odd ($m = 2k + 1$), save one sine factor and use $\sin^2 x = 1 - \cos^2 x$ to express the remaining factors in terms of cosine:

$$\int \sin^{2k+1} x \cos^n x\, dx = \int (\sin^2 x)^k \cos^n x \sin x\, dx$$
$$= \int (1 - \cos^2 x)^k \cos^n x \sin x\, dx$$

Then substitute $u = \cos x$. [Note that if the powers of both sine and cosine are odd, either (a) or (b) can be used.]

(c) If the powers of both sine and cosine are even, use the half-angle identities

$$\sin^2 x = \tfrac{1}{2}(1 - \cos 2x) \qquad \cos^2 x = \tfrac{1}{2}(1 + \cos 2x)$$

It is sometimes helpful to use the identity

$$\sin x \cos x = \tfrac{1}{2} \sin 2x$$

Strategy for Evaluating $\int \tan^m x \sec^n x\, dx$

(a) If the power of secant is even ($n = 2k$, $k \geqslant 2$), save a factor of $\sec^2 x$ and use $\sec^2 x = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$:

$$\int \tan^m x \sec^{2k} x\, dx = \int \tan^m x\, (\sec^2 x)^{k-1} \sec^2 x\, dx$$
$$= \int \tan^m x\, (1 + \tan^2 x)^{k-1} \sec^2 x\, dx$$

Then substitute $u = \tan x$.

(b) If the power of tangent is odd ($m = 2k + 1$), save a factor of $\sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$ to express the remaining factors in terms of $\sec x$:

$$\int \tan^{2k+1} x \sec^n x\, dx = \int (\tan^2 x)^k \sec^{n-1} x \sec x \tan x\, dx$$
$$= \int (\sec^2 x - 1)^k \sec^{n-1} x \sec x \tan x\, dx$$

Then substitute $u = \sec x$.

Exercises Page 484 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29 (another answer: $\frac{1}{8}\sec^8 x - \frac{1}{6}\sec^6 x + C$), 33, 39, 41, 45, 47, 61, 63.

7.3 Trigonometric Substitutions

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Expression	Substitution	Domain	Identity
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$0 \le \theta \le \pi,\ \theta \ne \frac{\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

$0 \le \theta < \frac{\pi}{2}$ or $\pi \le \theta < \frac{3\pi}{2}$

Example 7.19. Find $\int \sqrt{a^2 - x^2}\,dx$.

let $x = a\sin\theta$

$dx = a\cos\theta\,d\theta$

$$\int \sqrt{(a)^2 - (x)^2}\,dx = \int \sqrt{a^2 - a^2\sin^2\theta}\; a\cos\theta\,d\theta$$

$$= \int \sqrt{a^2(1-\sin^2\theta)}\; a\cos\theta\,d\theta$$

$$= \int a\sqrt{\cos^2\theta}\; a\cos\theta\,d\theta = \int a^2\cos^2\theta\,d\theta = a^2\int \cos^2\theta\,d\theta$$

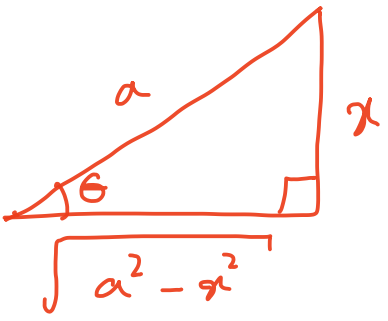
$$= a^2\int \frac{1+\cos 2\theta}{2}\,d\theta = a^2\int\left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right)d\theta$$

$$= \frac{a^2}{2}\theta + \frac{a^2}{2}\int \cos 2\theta\,d\theta = \frac{a^2}{2}\theta + \frac{a^2}{2}\left(\frac{1}{2}\right)\sin 2\theta + C$$

$$= \frac{a^2}{2}\theta + \frac{a^2}{4}\sin 2\theta + C$$

$x = a\sin\theta \iff \frac{x}{a} = \sin\theta \iff \theta = \sin^{-1}\frac{x}{a}$

$\sin\theta = \frac{x}{a} = \frac{\text{opposite}}{\text{hypotenuse}}$



$\sin 2\theta = 2\sin\theta\cos\theta$

$$\int \sqrt{a^2 - x^2}\, dx = \frac{a^2}{2}\theta + \frac{a^2}{4}\sin 2\theta + C$$

$$= \frac{a^2}{2}\theta + \frac{a^2}{4}(2)\sin\theta\cos\theta + C$$

$$= \frac{a^2}{2}\sin^{-1}\frac{x}{a} + \frac{a^2}{2}\left(\frac{x}{a}\right)\left(\frac{\sqrt{a^2 - x^2}}{a}\right) + C$$

$$= \frac{a^2}{2}\sin^{-1}\frac{x}{a} + \frac{x}{2}\sqrt{a^2 - x^2} + C$$

Example 7.20. Find $\int \frac{dx}{\sqrt{9+x^2}}$. $= \int \frac{dx}{\sqrt{(3)^2+(x)^2}}$

let $x = 3\tan\theta$

$dx = 3\sec^2\theta\, d\theta$

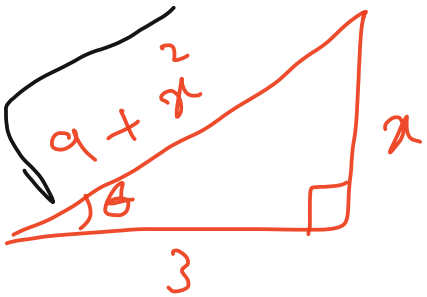
$= \int \frac{3\sec^2\theta\, d\theta}{\sqrt{9+9\tan^2\theta}} = \int \frac{3\sec^2\theta\, d\theta}{3\sqrt{1+\tan^2\theta}}$

$\int \frac{dx}{\sqrt{9+x^2}} = \int \frac{\sec^2\theta\, d\theta}{\sqrt{\sec^2\theta}} = \int \frac{\sec^2\theta}{\sec\theta}\, d\theta = \int \sec\theta\, d\theta$

$= \ln|\sec\theta + \tan\theta| + C = \ln\left|\frac{1}{\cos\theta} + \tan\theta\right| + C$

Example 7.21. Find $\int_2^4 \frac{\sqrt{x^2-4}}{x}dx$.

$\therefore x = 3\tan\theta \Leftrightarrow \tan\theta = \frac{x}{3} = \frac{\text{OPP}}{\text{Adj}}$



$\int \frac{dx}{\sqrt{9+x^2}} = \ln\left|\frac{1}{\frac{\sqrt{9+x^2}}{3}} + \frac{x}{3}\right| + C = \ln\left|\frac{\sqrt{9+x^2}}{3} + \frac{x}{3}\right| + C$

Example 7.21: Find $\int_2^4 \frac{\sqrt{x^2-4}}{x}\,dx = \int_2^4 \frac{\sqrt{(x)^2-(2)^2}}{x}\,dx$

let $x = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\,d\theta$

When $x=2 \Leftrightarrow 2 = 2\sec\theta$

$\sec\theta = 1 \Leftrightarrow \frac{1}{\cos\theta} = 1$

$\cos\theta = 1 \Leftrightarrow \theta = \cos^{-1}1 = 0$

When $x = 4 \Leftrightarrow 4 = 2\sec\theta$

$\sec\theta = 2 \Leftrightarrow \frac{1}{\cos\theta} = 2$

$\cos\theta = \frac{1}{2} \Leftrightarrow \theta = \cos^{-1}\frac{1}{2} = \frac{\pi}{3}$

$$= \int_0^{\frac{\pi}{3}} \frac{\sqrt{4\sec^2\theta - 4}}{2\sec\theta}(2\sec\theta\tan\theta\,d\theta)$$

$$= \int_0^{\frac{\pi}{3}} 2\sqrt{\sec^2\theta - 1}\,\tan\theta\,d\theta$$

$$= \int_0^{\frac{\pi}{3}} 2\sqrt{\tan^2\theta}\,\tan\theta\,d\theta$$

$$= 2\int_0^{\frac{\pi}{3}} \tan^2\theta\,d\theta$$

$$= 2\int_0^{\frac{\pi}{3}} (\sec^2\theta - 1)\,d\theta$$

$$= 2\left[\tan\theta - \theta\right]_0^{\frac{\pi}{3}} = 2\left[\left(\tan\frac{\pi}{3} - \frac{\pi}{3}\right) - (\tan 0 - 0)\right]$$

$$= 2\sqrt{3} - \frac{2\pi}{3} \approx 1.3697$$

Example 7.22. Find $\int \frac{dx}{\sqrt{x^2+2x+26}}$.

Example 7.23. Find $\int \frac{1}{x^2\sqrt{x^2+4}}dx$. $= \int \frac{dx}{x^2\sqrt{(x)^2+(2)^2}}$

let $x = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$

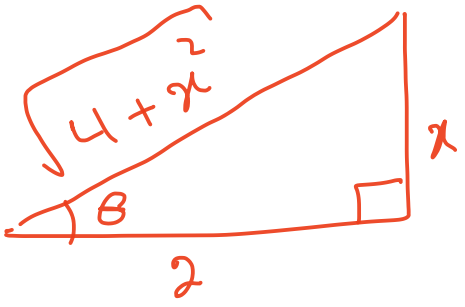
$$= \int \frac{2\sec^2\theta\, d\theta}{4\tan^2\theta\sqrt{4\tan^2\theta+4}}$$

$$= \int \frac{\sec^2\theta\, d\theta}{2\tan^2\theta(2)\sqrt{\tan^2\theta+1}} = \int \frac{\sec^2\theta\, d\theta}{4\tan^2\theta\sqrt{\sec^2\theta}} = \int \frac{\sec^2\theta\, d\theta}{4\tan^2\theta\sec\theta}$$

$$= \frac{1}{4}\int \frac{\sec\theta}{\tan^2\theta}d\theta = \frac{1}{4}\int \frac{1}{\cos\theta}\frac{(\cos^2\theta)}{\sin^2\theta}d\theta = \frac{1}{4}\int \frac{\cos\theta}{\sin^2\theta}d\theta$$

$$= \frac{1}{4}\int \frac{\cos\theta}{\sin\theta}\frac{1}{\sin\theta}d\theta = \frac{1}{4}\int \cot\theta\csc\theta\, d\theta = \frac{-1}{4}\csc\theta + C$$

$\because x = 2\tan\theta \iff \tan\theta = \frac{x}{2} = \frac{\text{Opp}}{\text{Adj}}$



$$\int \frac{1}{x^2\sqrt{x^2+4}}\,dx = \frac{-1}{4}\csc\theta + C = \frac{-1}{4}\cdot\frac{1}{\sin\theta} + C$$

$$= \frac{-1}{4}\cdot\frac{1}{\frac{x}{\sqrt{4+x^2}}} + C = \frac{-\sqrt{4+x^2}}{4x} + C$$

H.W

Example 7.24. Find $\int \frac{1}{\sqrt{x^2-a^2}}dx.$ $= \ln\left|\frac{x}{a} + \frac{\sqrt{x^2-a^2}}{a}\right| + C$

H.W (Trigo-substitution)

Example 7.25. Find $\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2+9)^{3/2}}dx.$ $= \int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{\sqrt{[(2x)^2+(3)^2]^3}}dx$

let $u = 4x^2+9 \Leftrightarrow x^2 = \frac{u-9}{4}$

$du = 8x\,dx$

when $x=0$, $u=9$

when $x = \frac{3\sqrt{3}}{2}$, $u=36$

$= \int_0^{\frac{3\sqrt{3}}{2}} \frac{x^2\, x\, dx}{(4x^2+9)^{\frac{3}{2}}}$

$= \int_9^{36} \frac{\frac{u-9}{4}\left(\frac{1}{8}\right)du}{u^{\frac{3}{2}}}$

$= \frac{1}{32}\int_9^{36} \frac{u-9}{u^{\frac{3}{2}}}du$ $= \frac{1}{32}\int_9^{36}\left(\frac{u}{u^{\frac{3}{2}}} - \frac{9}{u^{\frac{3}{2}}}\right)du$

$= \frac{1}{32}\int_9^{36}\left(u^{-\frac{1}{2}} - 9u^{-\frac{3}{2}}\right)du = \frac{1}{32}\left[2u^{\frac{1}{2}} + 18u^{-\frac{1}{2}}\right]_9^{36} = \cdots = \frac{3}{32}$

Example 7.26. Find $\int \frac{x}{\sqrt{3-2x-x^2}}dx.$

Exercises Page 491 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 23, 25, 27, 29, 37.

$\frac{P(x)}{Q(x)}$ such that $P(x)$ and $Q(x)$ are polynomials

7.4 Integration of Rational Functions Using Partial Fractions

Example 7.27. Find $\int \frac{2}{(x+1)^3}dx$.

let $u = x+1$
$du = dx$

$= \int \frac{2}{u^3}\,du = 2\int u^{-3}\,du = 2\,\frac{u^{-2}}{-2} + C$

$= -(x+1)^{-2} + C$

Note: $\frac{P(x)}{Q(x)}$, if $\text{Deg}\,P(x) \geq \text{Deg}\,Q(x)$, do long division first.

Example 7.28. Find $\int \frac{x^3+x}{x-1}dx$.

$= \int \left(x^2 + x + 2 + \frac{2}{x-1}\right) dx$

$= \frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x + 2\int \frac{dx}{x-1}$

let $u = x-1$
$du = dx$

$x^2 + x + 2$
$x-1 \;\overline{)\; x^3 + x}$
$-x^3 \pm x^2$
$x^2 + x$
$-x^2 \pm x$
$2x$
$-2x \pm 2$
2

Example 7.29. Find $\int \frac{2x+2}{x^2-4x+8}dx$.

$\int \frac{x^3+x}{x-1}dx = \frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x + 2\int \frac{du}{u}$

$= \frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x + 2\ln|u| + C$

$= \frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x + 2\ln|x-1| + C$

Example 7.30. Decompose $\dfrac{3x-1}{x^2-x-6}$ into partial fractions and find $\displaystyle\int \frac{3x-1}{x^2-x-6}dx$.

Example 7.31. Find $\int \dfrac{5x+3}{x^3-2x^2-3x}dx.$

Example 7.32. Find $\int \frac{x}{(x-3)^2} dx$.

Example 7.33. Find $\displaystyle\int \frac{3x^2 - 8x + 13}{(x+3)(x-1)^2} dx.$

Example 7.34. Find $\int \dfrac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx.$

Example 7.35. Find $\int \dfrac{6x^2 - 3x + 1}{(4x+1)(x^2+1)} dx.$

Example 7.36. Write out the form of the partial fraction decomposition of the function

$$\frac{x^3+x^2+1}{x(x-1)(x^2+x+1)(x^2+1)^3}$$

Example 7.37. Evaluate $\displaystyle\int \frac{\sqrt{x+4}}{x}dx$.

Exercises Page 501 1, 3, 5, 7, 11, 15, 17, 19, 21, 23, 27, 29, 31, 33, 35, 37, 39, 41, 47, 49, 51, 65.

7.5 Strategies for Integration

- **Standard Integral Formulas**

1. $\int u^r du = \dfrac{u^{r+1}}{r+1} \quad (r \neq -1)$
2. $\int \dfrac{1}{u} du = \ln|u| + C$
3. $\int e^u du = e^u + C$
4. $\int a^u du = \dfrac{a^u}{\ln a} + C \quad (a > 0,\ a \neq -1)$
5. $\int \sin u du = -\cos u + C$
6. $\int \cos u du = \sin u + C$
7. $\int \sec^2 u du = \tan u + C$
8. $\int \csc^2 u du = -\cot u + C$
9. $\int \sec u \tan u du = \sec u + C$
10. $\int \csc u \cot u du = -\csc u + C$
11. $\int \tan u du = \ln|\sec u| + C$
12. $\int \cot u du = \ln|\sin u| + C$
13. $\int \sec u du = \ln|\sec u + \tan u| + C$
14. $\int \csc u du = -\ln|\csc u + \cot u| + C$
15. $\int \dfrac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1}\dfrac{u}{a} + C$
16. $\int \dfrac{1}{a^2 + u^2} du = \dfrac{1}{a}\tan^{-1}\dfrac{u}{a} + C$
17. $\int \sinh u du = \cosh u + C$
18. $\int \cosh u du = \sinh u + C$

Example 7.38. Evaluate $\displaystyle\int \frac{\tan^3 x}{\cos^3 x} dx$.

Example 7.39. Evaluate $\int e^{\sqrt{x}} dx$.

Example 7.40. Evaluate $\int \frac{dx}{x\sqrt{\ln x}}$.

Example 7.41. Evaluate $\int \sqrt{\frac{1-x}{1+x}}dx$.

Exercises Page 507 1, 7, 19, 21, 31, 45, 57, 61, 63, 71, 75.

7.7 Numerical Integration

- Some indefinite integrals such as

$$\int \sin(x^2)dx, \quad \int \sqrt{1-x^4}dx, \quad \int \frac{\sin x}{x}dx$$

cannot be expressed algebraically in terms of elementary functions.

- To evaluate the definite integrals

$$\int_a^b \sin(x^2)dx, \quad \int_a^b \sqrt{1-x^4}dx, \quad \int_a^b \frac{\sin x}{x}dx$$

we need to rely on **numerical methods**.

- A few ways to approximate such definite integrals are:

1. **Riemann Sums**: left endpoints, right endpoints and midpoints.
2. **Trapezoidal Rule**
3. **Simpson's Rule**

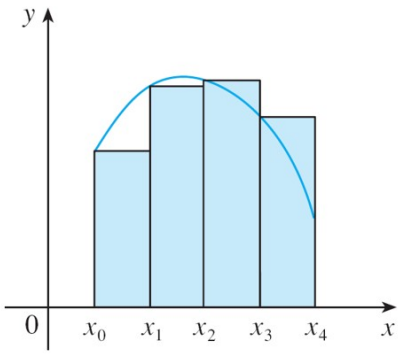
- **Riemann Sums**

$$\Delta x = \frac{b-a}{n}$$

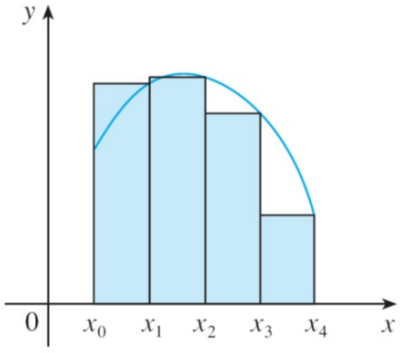
$$x_i = a + i\Delta x$$

$$\int_a^b f(x)dx \approx \sum_{i=1}^{n} f\left(\bar{x}_i\right)\Delta x$$

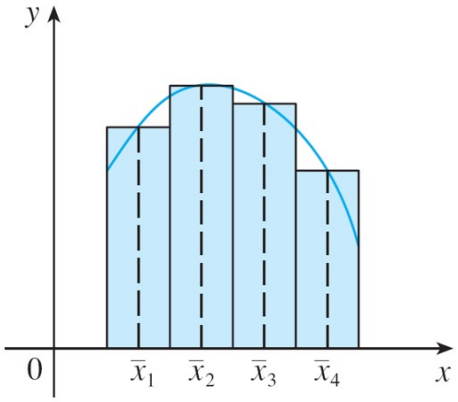
- **Left Riemann Sum** $\bar{x}_i = x_{i-1}$
- **Right Riemann Sum** $\bar{x}_i = x_i$
- **Midpoint Riemann Sum** $\bar{x}_i = \dfrac{x_{i-1}+x_i}{2}$



Left endpoint approximation



Right endpoint approximation



Midpoint approximation

Example 7.42. Approximate the definite integral $\int_1^3 \sqrt{4-x}dx$ using left, right, and midpoint Riemann sums with $n = 4$.

$\Delta x =$

i	x_i	$f(x_i) = \sqrt{4 - x_i}$
0		
1		
2		
3		
4		

$L_4 =$

$R_4 =$

i	$\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$	$f(\bar{x}_i) = \sqrt{4 - \bar{x}_i}$
1		
2		
3		
4		

$M_4 =$

Example 7.43. Approximate the definite integral $\int_0^2 \sin x^2 dx$ using a right Riemann sum with $n = 8$.

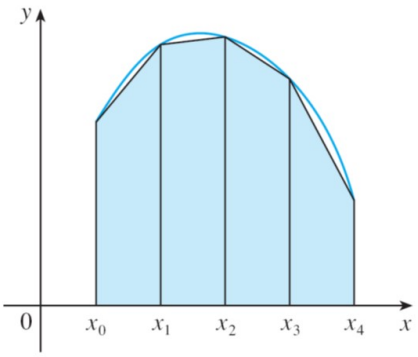
$\Delta x =$

i	x_i	$f(x_i) = \sin x_i^2$
0		
1		
2		
3		
4		
5		
6		
7		
8		

$R_8 =$

- **Trapezoidal Rule**

$$\int_a^b f(x)dx \approx \frac{\Delta x}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \ldots + 2f(x_{n-1}) + f(x_n)\right]$$



Example 7.44. Approximate the definite integral $\int_0^2 \sin x^2 dx$ using the trapezoidal rule with $n = 8$.

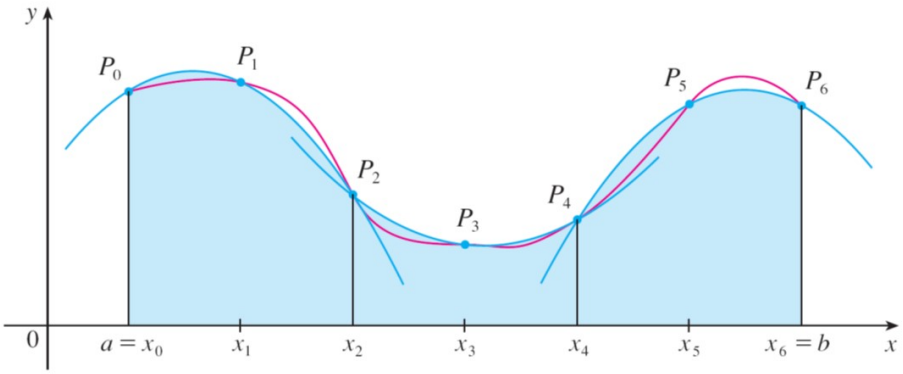
$\Delta x =$

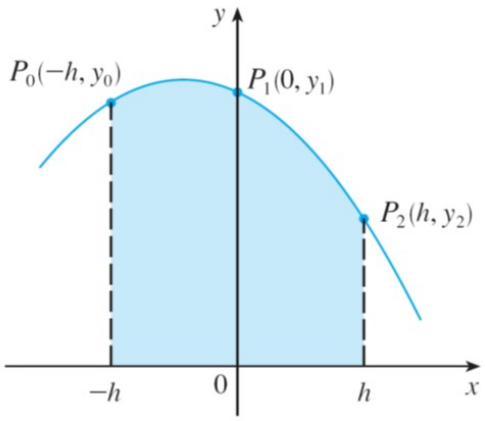
i	x_i	$f(x_i) = \sin x_i^2$
0		
1		
2		
3		
4		
5		
6		
7		
8		

$T_8 =$

- **Simpson's Rule**

$$\int_a^b f(x)dx \approx \frac{\Delta x}{3}\left[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \ldots + 4f(x_{n-1}) + f(x_n)\right]$$





Example 7.45. Approximate the definite integral $\int_0^2 \sin x^2 dx$ using the Simpson's rule with $n = 8$.

$\Delta x =$

i	x_i	$f(x_i) = \sin x_i^2$
0		
1		
2		
3		
4		
5		
6		
7		
8		

$S_8 =$

Exercises Page 524 7, 21(a), 31(a).

7.8 Improper Integrals

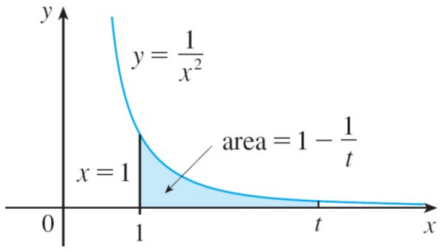
Infinite Intervals

- Integrals of the form

$$\int_0^\infty \frac{1}{1+x^2}dx, \quad \int_{-\infty}^{-1} xe^{-x^2}dx \quad \int_{-\infty}^{\infty} x^2e^{-x^2}dx$$

  are called **improper integrals** with infinite limits.

- Consider $\int_1^\infty \frac{1}{x^2}dx$.



- **Definition**

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx,$$
$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$$

If the limits on the right exists and have finite values, we say that the corresponding improper integrals **converge** and have those values. Otherwise, we say that the improper integrals **diverge**.

Example 7.46. Find, if possible, $\int_{-\infty}^{-1} xe^{-x^2}dx$.

Example 7.47. Find, if possible, $\int_{0}^{\infty} \sin x dx$.

Example 7.48. For what values of p does the integral $\int_1^\infty \frac{1}{x^p} dx$ converge?

Both Limits Infinite

- **Definition**

If both $\int_{-\infty}^{a} f(x)dx$ and $\int_{a}^{\infty} f(x)dx$ converge, then $\int_{-\infty}^{\infty} f(x)dx$ is said to converge and have value

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{a} f(x)dx + \int_{a}^{\infty} f(x)dx$$

Otherwise, $\int_{-\infty}^{\infty} f(x)dx$ diverges.

Example 7.49. Determine whether $\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx$ is convergent. If yes, find its value.

Example 7.50. Determine whether $\int_{-\infty}^{\infty} \frac{x}{1+x^2} dx$ is convergent. If yes, find its value.

Discontinuous Integrands

Example 7.51. Evaluate $\displaystyle\int_{-2}^{1} \frac{1}{x^2} dx$.

Integrands Infinite at an End Point

- **Definition**

Let f be continuous on the half-open interval $[a, b)$ and suppose that $\lim_{x \to b^-} |f(x)| = \infty$, then

$$\int_a^b f(x)\, dx = \lim_{t \to b^-} \int_a^t f(x)\, dx$$

provided that this limit exists and is finite, in which case we say that the integral converges. Otherwise, we say that the integral diverges.

Let f be continuous on the half-open interval $(a, b]$ and suppose that $\lim_{x \to a^+} |f(x)| = \infty$, then

$$\int_a^b f(x)\, dx = \lim_{t \to a^+} \int_t^b f(x)\, dx$$

provided that this limit exists and is finite, in which case we say that the integral converges. Otherwise, we say that the integral diverges.

Example 7.52. Evaluate, if possible, the improper integral $\int_0^2 \frac{dx}{\sqrt{4-x^2}}$.

Example 7.53. Evaluate, if possible, $\int_0^{16} \frac{1}{\sqrt[4]{x}} dx$.

Example 7.54. Evaluate, if possible, $\int_0^1 \frac{1}{x} dx$.

Example 7.55. For what values of p does the integral $\int_0^1 \frac{1}{x^p} dx$ converge?

Example 7.56. Evaluate, if possible, $\int_0^1 \ln x dx$.

Integrands Infinite at an Interior Point

- **Definition**

Let f be continuous on $[a, b]$ except at a number $c \in (a, b)$, and suppose that $\lim_{x \to c} |f(x)| = \infty$. Then we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)dx$$

provided both integrals on the right converge. Otherwise, we say that $\int_a^b f(x)dx$ diverges.

Example 7.57. Show that $\int_{-2}^{1} \frac{1}{x^2} dx$ diverges.

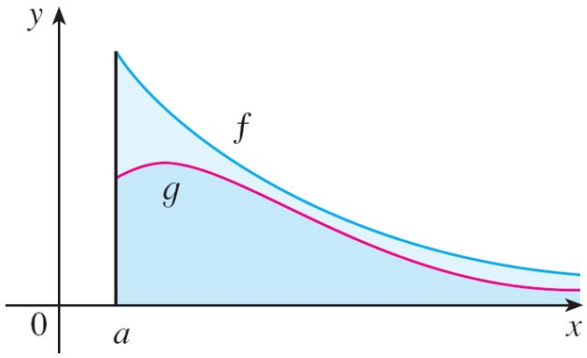
Example 7.58. Evaluate, if possible, the improper integral $\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$.

- **Comparison Test for Improper Integrals**

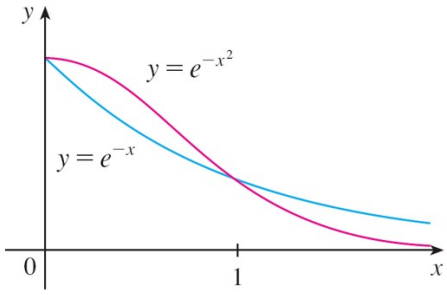
Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ when $x \geq a$.

(a) If $\int_a^\infty f(x)dx$ is convergent, so is $\int_a^\infty g(x)dx$.

(b) If $\int_a^\infty g(x)dx$ is divergent, so is $\int_a^\infty f(x)dx$.



Example 7.59. Determine whether $\int_0^\infty e^{-x^2}dx$ is convergent or divergent.



Example 7.60. Determine whether $\displaystyle\int_1^\infty \frac{1+e^{-x}}{x}dx$ is convergent or divergent.

Exercises Page 534 5, 7, 9, 11, 13, 15, 17, 19, 23, 25, 31, 33, 37, 39, 49, 51, 53, 55, 59.